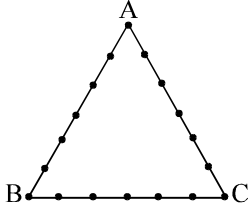


DPP NO.-4

- Q.1 A telegraph has x arms & each arm is capable of $(x - 1)$ distinct positions, including the position of rest. The total number of signals that can be made is _____.
- Q.2 The interior angles of a regular polygon measure 150° each. The number of diagonals of the polygon is
(A) 35 (B) 44 (C) 54 (D) 78
- Q.3 Number of different natural numbers which are smaller than two hundred million & using only the digits 1 or 2 is :
(A) $(3) \cdot 2^8 - 2$ (B) $(3) \cdot 2^8 - 1$ (C) $2(2^9 - 1)$ (D) none
- Q.4 5 Indian & 5 American couples meet at a party & shake hands. If no wife shakes hands with her own husband & no Indian wife shakes hands with a male, then the number of hand shakes that takes place in the party is :
(A) 95 (B) 110 (C) 135 (D) 150
- Q.5 The number of n digit numbers which consists of the digits 1 & 2 only if each digit is to be used atleast once, is equal to 510 then n is equal to:
(A) 7 (B) 8 (C) 9 (D) 10
- Q.6 Number of six digit numbers which have 3 digits even & 3 digits odd, if each digit is to be used atmost once is _____.
- Q.7 The tamer of wild animals has to bring one by one 5 lions & 4 tigers to the circus arena. The number of ways this can be done if no two tigers immediately follow each other is _____.
- Q.8 18 points are indicated on the perimeter of a triangle ABC (see figure). How many triangles are there with vertices at these points?
(A) 331 (B) 408
(C) 710 (D) 711
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- Q.9 An English school and a Vernacular school are both under one superintendent. Suppose that the superintendentship, the four teachership of English and Vernacular school each, are vacant, if there be altogether 11 candidates for the appointments, 3 of whom apply exclusively for the superintendentship and 2 exclusively for the appointment in the English school, the number of ways in which the different appointments can be disposed of is :
(A) 4320 (B) 268 (C) 1080 (D) 25920
- Q.10 A committee of 5 is to be chosen from a group of 9 people. Number of ways in which it can be formed if two particular persons either serve together or not at all and two other particular persons refuse to serve with each other, is
(A) 41 (B) 36 (C) 47 (D) 76
- Q.11 A question paper on mathematics consists of twelve questions divided into three parts A, B and C, each containing four questions. In how many ways can an examinee answer five questions, selecting atleast one from each part.
(A) 624 (B) 208 (C) 2304 (D) none
- Q.12 If m denotes the number of 5 digit numbers if each successive digits are in their descending order of magnitude and n is the corresponding figure, when the digits are in their ascending order of magnitude then $(m - n)$ has the value
(A) ${}^{10}C_4$ (B) 9C_5 (C) ${}^{10}C_3$ (D) 9C_3